**Project Summary**

Harmonizing FreeMoCap Movement Data for Rehabilitation Research

Free and open-source motion capture can expand access to quantitative movement measurements for rehabilitation researchers and investigators working in clinical settings. However, wider access to data collection does not ensure that the resulting measurements can be understood and reused across studies. Differences in anatomical definitions, pose-estimation methods, coordinate conventions, and processing can affect interpretation. These distinctions must remain available when data leave the software that produced them.

This Data STACHE project will harmonize existing FreeMoCap gait and balance data with the DAPR framework and build that harmonization into a reusable FreeMoCap export. The work will connect movement time series with study context, measurement definitions, and processing history. Existing standards, including Motion-BIDS, will inform the design. Consultation with the DAPR Resource Core will guide terminology mappings and proposals for information not adequately represented in current resources.

Aim 1 will harmonize the gait and balance datasets from our existing validation study, preserving the context needed to interpret measurements from different pose-estimation backends and the marker-based reference. The resulting data packages, dictionary, and mappings will establish the requirements for Aim 2. Aim 2 will implement metadata capture and storage within the supported FreeMoCap workflow and an export capability that carries this information with the numerical data. Evaluation will check that exports reproduce the established representation, preserve measurements, and support representative reuse examples.

Within 12 months, the project will deliver harmonized gait and balance datasets, documented export software, reviewed mappings, and an evaluation report with worked examples. These outputs will support a manuscript submission plan and follow-on rehabilitation data research. By making the same export capability available to other FreeMoCap researchers, the pilot will provide a practical foundation for future harmonization across studies. It uses existing data and requires no new recruitment.

**Project Narrative**

Affordable, open-source motion capture can make quantitative movement measurements accessible to more rehabilitation researchers, including investigators working in clinical settings. This project will harmonize existing FreeMoCap gait and balance data with shared terminology and the information needed to understand how measurements were obtained. Incorporating that harmonization into a reusable FreeMoCap export will help researchers prepare interpretable movement data for sharing and future use across studies.

## Specific Aims

Harmonizing FreeMoCap Movement Data for Rehabilitation Research

Movement measurements are central to understanding walking and balance, yet their reuse across studies is limited by differences in data organization and measurement definitions. A variable called knee angle may reflect different landmarks, segment axes, or rotation sequences. Likewise, a trajectory may represent a surface marker, an estimated joint center, or a pose-estimation landmark. Pooling these measurements without their context can obscure physiological differences or mistake measurement differences for biological variation. Accessible systems such as FreeMoCap increase opportunities to collect movement data, making interoperable outputs and explicit provenance increasingly important.

FreeMoCap is a free and open-source markerless motion capture system that produces three-dimensional full-body kinematic data from consumer cameras (1), with over 10,000 GitHub stars and at least 15,000 users across 153 countries. Published applications span reaching and grasping biomechanics (2), gait analysis (3), and robotic exoskeleton design and verification (4). Independent studies have evaluated FreeMoCap against marker-based motion capture for selected tasks and outcomes (5,6).

This breadth of use creates an opportunity to make harmonized outputs available to researchers at the point of export. Our own validation work has characterized gait and balance measurements across multiple pose-estimation backends through comparison with marker-based measurements (7). This experience highlights the importance of preserving measurement definitions and processing history alongside numerical data, and provides existing datasets and analyses with which to develop and evaluate a shared representation.

The objective of this Data STACHE pilot is to harmonize existing gait and balance data with the DAPR framework and incorporate that harmonization into a reusable FreeMoCap export. We will work with the Resource Core to map relevant information to existing terminology and identify metadata needs for movement time series. Existing resources, including Motion-BIDS, will inform the representation and organization of the data. Harmonization of the validation datasets will establish the requirements for software that other researchers can use with supported FreeMoCap recordings.

**Aim 1 Harmonize existing FreeMoCap gait and balance validation datasets.** We will inventory the study's movement measurements and the information needed to interpret them, including task context, measurement definitions, and acquisition and processing provenance. With DAPR consultation, we will establish a data dictionary and terminology mappings, draw on applicable standards, and document unresolved metadata needs. We will assemble harmonized datasets and evaluate preservation of numerical values and measurement meaning. The outcome will be documented gait and balance data packages that establish the representation and metadata requirements for the export.

**Aim 2 Build the harmonized representation into a reusable FreeMoCap export.** We will implement the architecture needed to capture and retain relevant metadata during processing, connect researcher-supplied study context to recordings, and export numerical data with their definitions and provenance. Using the Aim 1 datasets as a reference, we will evaluate exports across the supported tasks and processing methods, check handling of missing metadata, and demonstrate representative reuse. The outcome will be tested export software and documentation that enable other researchers to prepare DAPR-aligned outputs from supported FreeMoCap recordings.

Within 12 months, the pilot will deliver harmonized gait and balance datasets, reviewed mappings, the export capability, and an evaluation report. Making this capability available through free and open-source software will establish a practical foundation for future multi-study harmonization. Findings will inform DAPR's work with movement time series and support plans for a methods manuscript and follow-on research.

## Research Strategy

### Significance and Preliminary Work

FreeMoCap makes quantitative movement measurements accessible through consumer cameras and free, open-source software (1). Its use across research settings creates an opportunity to improve how movement data are documented and shared. For rehabilitation research, reuse requires information about the protocol, measurement definitions, and processing methods; familiar variable names alone do not establish comparability (8).

Analysis choices can affect the measurements being shared. For example, in gait, anthropometric model selection can alter estimated mediolateral and vertical center-of-mass motion (9). In a 2025 ISB community consultation, 46% of respondents identified insufficient methodological detail as the biggest obstacle to comparing joint-motion data across research groups (10). Preserving these details alongside numerical outputs would help researchers assess which measurements are suitable for reuse and comparison.

Our gait and balance validation compares FreeMoCap outputs from MediaPipe, RTMPose, and ViTPose with Qualisys marker-based measurements (7). The gait comparisons in Figure 1 illustrate how pose-estimation choices can affect derived movement measurements. Developing this validation workflow has also required us to specify anatomical definitions and joint-angle conventions that are not conveyed by the resulting numerical arrays. This experience provides a practical basis for identifying the metadata that should accompany shared movement data and for evaluating whether harmonized exports preserve that information.

DAPR's development priorities include infrastructure for gait time series. [CITE: DAPR] This pilot will connect FreeMoCap outputs with DAPR's terminology and mapping resources, informed by applicable Motion-BIDS conventions (8). Harmonizing the existing data will establish metadata requirements; incorporating them into FreeMoCap will help other researchers retain that information during processing and export.

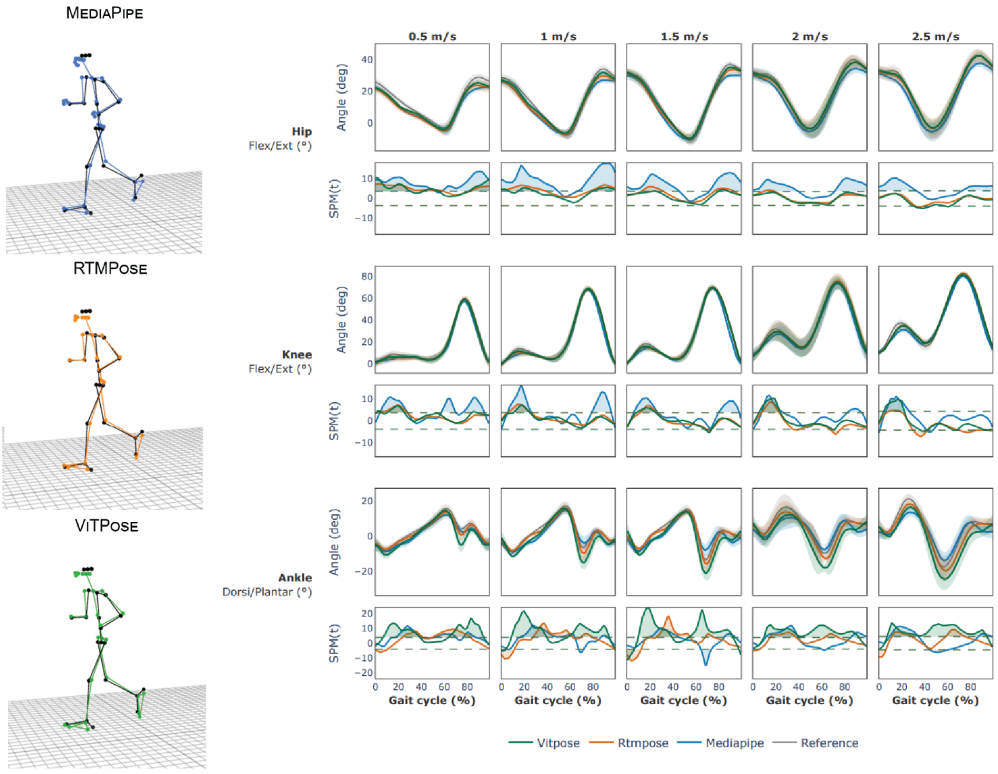



Figure 1: **Existing FreeMoCap gait validation.** Left: example reconstructions from three pose-estimation backends. Right: sagittal hip, knee, and ankle kinematics across five treadmill speeds (0.5-2.5 m/s), with Qualisys as the gray reference. Lower rows show statistical parametric mapping (SPM) t-statistics for comparisons with the reference. These comparisons motivate preserving tracking methods, measurement definitions, and processing history. [CITE: Validation]

## Approach

### Study Design and Existing Data

The project will use the existing FreeMoCap validation release, comprising six participants and 24 treadmill gait and standing balance trials, with aligned trajectories from MediaPipe, RTMPose, ViTPose, and Qualisys. Their different tasks and processing methods will support development and evaluation of a common representation; the reusable export will provide the route to future multi-study use. The scope includes reconstructed trajectories, available kinematic time series and summaries, and associated metadata. No new recruitment, video reconstruction, or pose-model training is required.

The validation study provides both shared movements observed through different measurement methods and distinct gait and balance task contexts. This combination allows us to evaluate whether a common representation preserves differences among methods while describing task-specific information consistently. Existing analysis code and configuration records will provide reference calculations and help identify metadata that FreeMoCap should retain systematically for future recordings.

### Aim 1: Harmonize Existing Gait and Balance Validation Datasets

In months 1-3, we will inventory the selected data and source records, document versions and permissions, and define representative reuse analyses. The dictionary will cover study and coded participant identifiers, sessions and trials, task conditions, measurement definitions, units, timing, coordinate frames, and acquisition and processing provenance. We will distinguish information available from the software workflow from context supplied through study records, and specify required and optional fields for each supported data type. Missing information will remain explicit.

Metadata requirements will reflect the measurements being shared. Joint-angle records will describe segment definitions, anatomical axes, sign conventions, and the rotation sequence or projection method, referencing applicable ISB recommendations and documenting the conventions actually used (11). For center-of-mass estimates, we will identify the anthropometric model and implemented parameters, including segment definitions, mass fractions, and relative segment COM locations (12). Other records will identify the pose estimator, model and software versions, filtering, interpolation, and analysis intervals. These examples will guide a common metadata core with additional fields for particular measurements. Definitions and parameters will be machine-readable and linked to the relevant numerical outputs.

With the DAPR Resource Core, we will review mappings to DAPR and established terminology, including NINDS or OMOP concepts where their definitions apply. Each mapping will retain the source definition, target identifier and version, and any transformation. Exact matches, matches requiring transformation, related concepts, and unresolved mappings will remain distinguishable. The PI and mentor will review mappings used in the demonstrations; missing concepts will be proposed to DAPR with definitions and examples. Motion-BIDS will inform applicable conventions for signals, channels, reference frames, and data organization, with versions, adopted conventions, and additional metadata documented explicitly.

Consultation with the Resource Core will begin by confirming the initial scope and receiving requirements. Intermediate review will address the dictionary, mapping decisions, and representative exports; final review will address the completed packages, remaining limitations, and handoff or import procedures. Meeting dates and the extent of Core involvement will be agreed with DAPR.

In months 3-5, we will assemble harmonized gait and balance packages that link numerical data, definitions, provenance, and source records through stable identifiers. Files will use documented, openly readable formats; Parquet is an option for numerical tables with linked metadata files. Original values and missingness will be preserved, and any unit or coordinate transformation will be explicit. Original timestamps, synchronized data, and waveforms normalized to a gait cycle will remain distinguishable.

Evaluation will account for each selected source field as represented or excluded with a reason. Coverage will be reported by task, data type, and measurement method, distinguishing available information, recovered information, and unresolved gaps. Checks will assess identifiers, source links, dimensions, units, timing, and missing-value masks. Unchanged arrays must match exactly after lossless serialization; tolerances for explicit transformations will be fixed before evaluation using source precision and documented test cases. Representative analyses will test whether the attached definitions and inputs support reuse. By month 5, the packages, dictionary, mappings, and initial coverage and fidelity report will establish the reference representation for Aim 2.

**Research Strategy Continued**

### Aim 2: Build the Harmonized Representation into a Reusable FreeMoCap Export

In months 5-9, we will incorporate the Aim 1 requirements into FreeMoCap's processing and export workflow. Recordings and derived measurements will be associated with versioned metadata records that retain relevant information through processing stages. Available software versions, pose-estimation settings, and processing parameters will be captured from supported workflows; measurement definitions and calculation settings will be linked to their outputs. Information recovered retrospectively in Aim 1 will identify what future recordings need to store systematically.

Researchers will supply context that software cannot infer, such as task conditions and protocol descriptions, through a documented recording or study manifest. The exporter will combine this context with stored processing metadata and produce harmonized numerical and metadata files. Checks will identify missing required fields, inconsistent descriptions, and unsupported data types. An export report will describe supported uses and unresolved gaps. Older recordings will retain missing provenance unless it can be recovered from a documented source.

In months 8-10, we will compare exports with the Aim 1 reference packages across supported gait and balance tasks and processing methods. Automated checks will assess numerical fidelity, identifiers, units, timing, measurement definitions, and provenance. Constructed test cases will evaluate metadata capture and propagation, researcher-supplied context, and handling of missing essential fields or incompatible definitions. Discrepancies will be investigated before release. Evaluation will distinguish file validity from sufficient documentation for a particular analysis.

Worked examples will reproduce selected analyses from exported data using documented environments and instructions. The PI will run them in a clean environment and compare results with native analyses using predefined tolerances. The mentor will review representative exports, instructions, and results for clarity and consistency with measurement definitions. A query example will select records by task and measurement descriptors while retaining method distinctions. Findings will guide revisions to the exporter and documentation, with release materials completed in months 10-12.

### Rigor, Feasibility, and Responsibilities

Results will be descriptive and reported by task and measurement method, retaining shared participant and trial identities. Versioned source files, mappings, transformation parameters, and software environments will support reproducibility. Aaron T. Cherian will lead harmonization, metadata architecture, implementation, evaluation, and DAPR coordination. Jonathan Samir Matthis, PhD, President and CEO of the FreeMoCap Foundation, will provide mentorship and review of scientific assumptions, mappings, software design, and findings. Monthly meetings and checkpoints at months 3, 6, 9, and 12 will track progress.

The initial release will support defined FreeMoCap versions, data types, and metadata requirements established from the validation datasets. Tested changes will be submitted through FreeMoCap's normal review process. A documented exporter will remain runnable from an archived release if inclusion in the general application takes longer. Optional metadata and graphical interface enhancements may be deferred to protect core functionality.

### Limitations and Alternatives

Evaluation within one validation study will establish performance for its supported data; transfer to independent cohorts and sites will require later assessment. Missing information and meaningful differences among pose estimators and anatomical models will remain visible. DAPR consultation will guide terminology and receiving requirements, with local definitions retained for unresolved terms. If central ingestion is unavailable, we will provide versioned packages, mappings, and a documented handoff. Harmonized representation does not establish clinical validity or make different measurements interchangeable.

**Dissemination Sharing and Future Research Plan**

The project will release harmonized gait and balance datasets, a data dictionary and terminology mappings, reusable FreeMoCap export software, and an evaluation report. Documentation will explain the supported inputs, metadata captured and stored during processing, study context supplied by the researcher, and known limitations. Worked examples will show how to prepare an export, inspect measurement definitions, select relevant records, and reproduce representative analyses.

By month 10, we will prepare a release candidate for DAPR review. By the end of the award, permitted data, metadata, code, and evaluation materials will be archived in an established repository such as Zenodo, with persistent identifiers and source attribution. GitHub will support development and issue reporting. Source licenses will be preserved, and software incorporated into FreeMoCap will follow its applicable licensing requirements. Scientific data will be shared by publication or, for other findings, by the award's end, consistent with the Data Management and Sharing Plan.

A short tutorial and an offered DAPR presentation will demonstrate the export and its use with the harmonized datasets. We will submit the integration changes for review through the FreeMoCap project's normal contribution process and provide a documented way to run the supported export independently of a particular application release. Feedback received through demonstrations and repository issues will inform future improvements. Archived releases and versioned documentation will preserve the pilot's results as the software evolves.

We plan to submit at least one peer-reviewed methods or research-software manuscript describing the representation and its evaluation. The paper will address the metadata needed to interpret gait and balance time series, how existing standards and DAPR terms cover those needs, and how metadata capture and export preserve that information for reuse. During the award, we will assemble results and a manuscript outline and identify a suitable journal. The anticipated submission window is the year following completion, allowing evaluation findings and coauthor review to shape the paper. Submission during the funded year and journal acceptance are not completion milestones.

The pilot will inform a larger study of movement-data sharing across rehabilitation cohorts and acquisition settings. In the latter half of the award, we plan to discuss mechanism fit with NICHD/NCMRR, identify potential clinical-research collaborators, and prepare follow-on specific aims. An NIH R21 or a mechanism recommended by program staff is the anticipated route. We will target an appropriate application cycle in the year after completion, subject to eligibility, an active funding opportunity, and partner readiness. The pilot's coverage results and documented metadata gaps will define the next research questions.

The PI will coordinate the six-month progress report, final scientific and expenditure reports, and required annual follow-up reporting. Records will distinguish completed outputs from planned submissions and proposed mappings from terms accepted by DAPR. The final report will identify the harmonized datasets, supported exports, evaluation results, and remaining work needed for broader adoption.

**Project Timeline**

The project runs for 12 months from the approved start date. Work will progress from harmonization of the existing gait and balance datasets to implementation and evaluation of the reusable FreeMoCap export. Consultation dates will be agreed with the DAPR Resource Core.

| Period | Activities | Milestone and deliverable |
|---|---|---|
| Months 1-2 | Complete institutional and data-use review; freeze source release; inventory gait and balance outputs and associated metadata. | Source manifest; documented permissions; initial scope of supported data and reuse examples. |
| Month 3 | Review the dictionary, existing standards, DAPR mappings, and metadata available from processing or study records. | Representation version 0.1; mapping decisions; required and optional fields; unresolved terms logged. |
| Months 4-5 | Assemble harmonized gait and balance packages; assess coverage, numerical fidelity, and representative reuse. | Aim 1 datasets, dictionary, mappings, and initial evaluation; requirements for metadata storage and export. |
| Month 6 | Implement recording and measurement metadata records and their links through processing; review progress. | Initial metadata architecture and capture mechanisms; six-month progress report and budget update. |
| Months 7-8 | Implement study-context input, export generation, and checks; compare initial exports with Aim 1 packages. | Runnable export; supported-input documentation; initial fidelity and metadata propagation results. |
| Month 9 | Test supported tasks and methods, missing metadata, and clean-environment reuse; conduct mentor review and assess DAPR handoff needs. | Reuse checks and mentor review log; revised software and documentation; follow-on concept outline. |
| Months 10-11 | Prepare release candidate, tutorial, evaluation report, and manuscript outline; discuss future funding. | DAPR review materials; organized results; draft follow-on aims and software contribution for review. |
| Month 12 | Archive permitted outputs; demonstrate the export; complete reporting. | Persistent identifiers for datasets, exporter, dictionary, mappings, tutorial, and final reports. |
| Year after completion | Planned manuscript submission and application to an appropriate follow-on opportunity. | Target windows refined through coauthor review, program consultation, eligibility, and partner readiness. |

The PI will carry out harmonization, software development, and evaluation, with monthly mentor review and quarterly checkpoints. Optional fields and graphical interface enhancements may be deferred to protect core functionality. Manuscript submission and acceptance are not month-12 completion criteria.

**Bibliography**


1. FreeMoCap: A Free, Open Source Markerless Motion Capture System [Internet]. 2026 Aug 27. doi:10.5281/zenodo.22131400
2. Catching Small Balls with a Power Grip and Large Balls with a Precision Grip: Distinct Roles for the Reach and Grasp. Experimental Brain Research. 2026 Jan 31;244(3):38. doi:10.1007/s00221-025-07223-4 PubMed PMID: 41619057.
3. Understanding Gait Alterations: Trunk Flexion and Its Effects on Walking Neuromechanics. Journal of Experimental Biology. 2024 Oct 11;227(19):jeb249307. doi:10.1242/jeb.249307
4. A Hybrid Verification Method for an Actuated Lower-Limb Exoskeleton Based on Mathematical and Simulation Analyses. Robotics. 2026 Sep ;15(9):166. doi:10.3390/robotics15090166
5. A Low-Cost Markerless Motion Capture System to Automate Functional Gait Assessment: Feasibility Study. PLOS ONE. 2026 Jun 2;21(6):e0346606. doi:10.1371/journal.pone.0346606
6. Validation of Free-to-Use, Markerless Motion Capture for Field Based Measurements of Sagittal Plane Kinematics [SSRN Scholarly Paper] [Internet]. Rochester, NY; 2025 [cited 2026 Mar 24]. Available from: https://papers.ssrn.com/abstract=5089509 doi:10.2139/ssrn.5089509
7. Open-Source Development and Validation of a Low-Cost Markerless System for Quantitative Motion Analysis [Internet]. 2026 [cited 2026 Oct 15]. Available from: https://repository.library.northeastern.edu/files/neu:ms425z58z
8. Motion-BIDS: An Extension to the Brain Imaging Data Structure to Organize Motion Data for Reproducible Research. Scientific Data. 2024 Jul 2;11(1):716. doi:10.1038/s41597-024-03559-8
9. Does the Anthropometric Model Influence Whole-Body Center of Mass Calculations in Gait?. Journal of Biomechanics. 2017 Jul 5;59:23–8. doi:10.1016/j.jbiomech.2017.05.007 PubMed PMID: 28549601.
10. Data Sharing and Kinematic Standardisation: Stockholm 2025 [Internet]. [cited 2026 Oct 15]. Available from: https://media.isbweb.org/news/isb-now/204-september-2025/1043-kinematic-standardisation?utm_source=chatgpt.com
11. ISB Recommendation on Definitions of Joint Coordinate System of Various Joints for the Reporting of Human Joint Motion–Part I: Ankle, Hip, and Spine. International Society of Biomechanics. Journal of Biomechanics. 2002 Apr ;35(4):543–8. doi:10.1016/s0021-9290(01)00222-6 PubMed PMID: 11934426.
12. Biomechanics and Motor Control of Human Movement. 4th ed. Hoboken, N.J: Wiley; 2009. 370 p.

## Resources

### Institutional Environment and Infrastructure

The FreeMoCap Foundation is a nonprofit organization responsible for the development and stewardship of FreeMoCap, an open-source platform for accessible markerless motion capture. The proposed work will make a harmonized export capability available for supported FreeMoCap outputs. Familiarity with the software and its development process provides a practical basis for implementing the workflow and documenting how researchers can use it.

Aaron T. Cherian, PhD, is the proposed PI and will carry out data harmonization, metadata architecture, export implementation, checks, and documentation. His experience with markerless motion capture, gait and balance analysis, and comparisons among tracking methods will inform metadata selection and measurement definitions. He will coordinate DAPR consultation, lead evaluation, and develop dissemination and follow-on research plans.

Jonathan Samir Matthis, PhD, President and CEO of the FreeMoCap Foundation, will serve as project mentor. He will review measurement definitions and mappings, advise on metadata architecture and export design, and review evaluation results and release documentation. The budget supports the PI and mentor. The PI will run reproducibility checks in a clean environment; mentor review will identify questions or gaps in the examples and documentation. The initial release will support defined FreeMoCap versions and data types, the existing gait and balance datasets, and representative analyses.

Development will use existing computers and the team's repository workflow, with modest funds for storage and computational evaluation. Existing reconstructed data provide the necessary inputs. The project does not require new recording equipment, recruitment, or a model-training cluster. Any additional essential institutional or collaborator resources will be documented explicitly before submission.

### Data and Computational Resources

The data source is the public FreeMoCap validation release and analysis repository. It provides aligned trajectories from multiple pose-estimation backends and a marker-based reference, together with gait and balance analysis code. The data and existing analyses will support harmonization in Aim 1 and provide reference packages for evaluating the exporter in Aim 2. Measurements from the same trial will retain their relationship and method identities. According to the repository, the public release excludes raw videos and direct participant identifiers; institutional review will separately consider investigator familiarity with participants.

The gait and balance datasets belong to the same validation study and provide differing task contexts and measurement types for the pilot. Associated configurations, analysis code, and study records will help recover available metadata and identify information that FreeMoCap should retain systematically. The final source inventory will document the release, eligible records, and permissions for use and sharing. Broader application across independent studies will be enabled by the reusable export and assessed in subsequent work.

The implementation will add persistent metadata records and links through the supported FreeMoCap processing workflow, using documented, openly readable numerical and metadata formats with example loading and analysis code. Parquet is an available option for numerical tables, with linked files for detailed metadata and terminology mappings. Existing analysis code will provide reference calculations where supported. The package definitions will be independent of any one internal software library so that other researchers can inspect and use the exported data.

Source files, working copies, metadata, and outputs will be connected through identifiers and checksums. Computation will focus on conversion, automated checks, and representative analyses using existing hardware and the budgeted allowance. Code, configurations, dependencies, and relevant source versions will be retained together. Public releases will be archived in an established repository; restricted working materials, if approved, will use institutional access controls.

**Resources Continued**

**DAPR Resources and Existing Standards**

We request Resource Core consultation to review the representation, map the selected study descriptors and measurements, and identify information that current terminology does not adequately capture. Initial consultation will address scope and receiving requirements. Intermediate review will examine the dictionary, mapping decisions, and representative exports. Final review will address study packages, limitations, and handoff or import procedures. Meeting dates and effort will be agreed with the Core.

DAPR describes a Schema, Dictionary, Thesaurus, and MAPR mapping application. Where appropriate, mapping tools may suggest candidate terms; the team will review definitions, units, and relationships before accepting a match. Broader or related concepts will remain distinguishable from exact mappings. Terms without an adequate match will retain clear definitions and be proposed for DAPR review.

Existing resources, including Motion-BIDS, will inform how motion signals, channels, reference frames, and related metadata are represented. Applicable conventions will be adopted and the scope of conformance documented. Additional metadata will remain explicit, with source links and a rationale for its inclusion. This will allow the export design to draw on established work while responding to the needs identified in the study inventory and DAPR consultation.

The contribution to DAPR will include harmonized gait and balance datasets, reviewed mappings, a reusable export workflow, and an evaluation of the information needed for interpretation and reuse. If a central receiving interface is unavailable during the award, we will provide versioned packages and a documented handoff.

Versioned releases and accompanying documentation will be archived in an established repository to preserve the pilot’s outputs and support their continued use. Future maintenance and expansion will follow the project’s established development processes.

**Resource Stewardship and Career Development**

The PI and mentor will use monthly reviews and quarterly checkpoints to assess progress against the project aims and timeline and address implementation or evaluation challenges. Matthis will provide scientific and technical guidance and support the PI’s development of manuscript materials and follow-on research aims. Collaboration with DAPR will develop the PI’s expertise in rehabilitation terminology, data standards, and harmonization across studies.

**Data Management and Sharing Plan**

**1.** Will there be maximum appropriate sharing of scientific data underlying peer-reviewed publications and other findings resulting from the work supported by this award (including preprints, refereed papers reported at conferences, and other findings)?

Yes.

**2.** Will the scientific data underlying peer-reviewed publications be shared by the time of publication, or for other findings, by the end of the period of performance, which includes no-cost extensions?

Yes.

**3.** Will shared scientific data be made available for at least as long as required by applicable data repository policies and/or journal policies?

Yes.

**4.** If you answered “no” to elements 1, 2, or 3, or if you anticipate that sharing will be limited in some other way, please describe these limitations and the ethical, legal, or technical factors for them. Your response should specify a particular reason(s) for limiting sharing.

No additional limitations to sharing the scientific data and associated metadata generated by this project are anticipated.

**5.** If scientific data derived from human research participants will be shared, will privacy, rights, and confidentiality of participants be protected as outlined in NOT-OD-22-213?

Yes.

Public releases will contain only permitted, nonidentifying numerical data and metadata. Restricted working materials, if approved, will remain access controlled and will not be deposited publicly.

**6.** In the table below, please list:

| Expected Data Type | Established Repository or Example |
|---|---|
| Reconstructed movement trajectories, kinematic time series and summaries, measurement definitions, processing history, and permitted study metadata. | Zenodo, linked to the source validation study; harmonized gait and balance packages and mappings provided to DAPR. |
| Data dictionary and mappings, evaluation results, metadata capture and export software, example loading and analysis code, and reproducible environments. | Zenodo archival releases; GitHub for maintained source code and documentation. |

**7.** For studies subject to the NIH Genomic Data Sharing Policy (GDS) (e.g., using NIH funds to generate large-scale human genomic data):

**a.** Will you share all large-scale human genomic and associated data in a NIH-designated repository according to the accelerated timelines expected in the GDS Policy?

Not Applicable. No genomic data are proposed.

**b.** Do you anticipate that when sharing you will be able to meet the expectations of the Institutional Certification in the GDS Policy?

Not Applicable. No genomic data are proposed.

**Human Subjects Determination**

**Project title:** Harmonizing FreeMoCap Movement Data for Rehabilitation Research
**Principal Investigator:** Aaron T. Cherian
**Applicant Organization:** FreeMoCap Foundation

**Proposed Activities and Data Source**

The project will use the publicly released FreeMoCap gait and balance validation dataset to develop and evaluate harmonized movement-data packages and a reusable FreeMoCap exporter. The participant-level inputs will consist of publicly available numerical trajectories, derived measurements, task descriptors, and associated study metadata. Software and methodological documentation will support the description of processing steps and measurement definitions.

The original dataset was collected under Northeastern University IRB approval. Northeastern University will not participate in the proposed project. No new participants will be recruited or contacted, no new recordings will be collected, and no intervention or clinical trial is proposed.

**Proposed Exemption**

For application purposes, the PI proposes that the project qualifies for Exemption 4 under 45 CFR 46.104(d)(4)(i), concerning secondary research using publicly available information. The data were collected for an earlier validation study and are now publicly available for reuse. The proposed work will reorganize and document those data, evaluate the preservation of measurements and metadata during export, and develop reusable software. No additional nonpublic participant-level information will be obtained or incorporated into the project.

This exemption designation represents the PI's preliminary assessment for the application. A formal determination for the proposed Foundation-based project has not yet been obtained.

**Identifiability and Privacy Protections**

The public dataset omits direct participant identifiers. The PI retains access to information that can connect participant codes to identities; accordingly, this justification does not rely on an assertion that the PI cannot identify participants.

Participant linkage keys, names, contact information, consent records, and identifiable source videos will not be included in the project's analytical workspace or public outputs. The project will not link the research data to participant identities or other private records. Data use and sharing will follow the permissions applicable to the source dataset.

**Confirmation Before Project Initiation**

The Foundation will coordinate with DAPR to identify an appropriate independent authority to confirm the proposed classification and obtain any required determination or approval before beginning the funded secondary data analyses. The human subjects documentation will be updated to reflect that determination. Any proposed expansion to nonpublic participant data or new participant interaction will be assessed before implementation.

**Detailed Budget for Initial Budget Period**

**Direct costs only**

Program Director/Principal Investigator: Aaron T. Cherian [proposed PI]

Period: [START DATE] through [END DATE]

Proposed budget assumptions require institutional approval; see Budget Justification.

List PERSONNEL (applicant organization only). Use calendar, academic, or summer months. Enter whole dollar amounts requested for salary and fringe benefits.

| Name | Role | Cal. months | Acad. months | Sum. months | Inst. base salary | Salary requested | Fringe benefits | Total |
|---|---|---|---|---|---|---|---|---|
| Aaron T. Cherian | PD/PI | 3.51 | | | 60,000 | 17,538 | 3,508 | 21,046 |
| Jonathan S. Matthis | Mentor | 1.17 | | | 80,000 | 7,795 | 1,559 | 9,354 |
| **Personnel subtotals** | | | | | | 25,333 | 5,067 | **30,400** |
| Consultant costs | | | | | | | | 0 |
| Equipment | | | | | | | | 0 |
| Supplies | | | | | | | | 0 |
| Travel | | | | | | | | 0 |
| Inpatient care costs | | | | | | | | 0 |
| Outpatient care costs | | | | | | | | 0 |
| Alterations and renovations | | | | | | | | 0 |
| Other expenses: project storage and backup $600 | | | | | | | | 600 |
| Consortium/contractual costs: direct costs | | | | | | | | 0 |
| **Subtotal direct costs for initial budget period** | | | | | | | | **$31,000** |
| Consortium/contractual costs: facilities and administrative costs | | | | | | | | 0 |
| **Total direct costs for initial budget period** | | | | | | | | **$31,000** |

Indirect costs are detailed in the Budget Justification. The requested indirect costs are $2,480, bringing the total request to $33,480.

## Budget Justification

Project period: 12 months from the approved start date.

Requested direct costs: $31,000. Requested indirect costs: $2,480. Total request: $33,480.

The budget supports harmonization of existing gait and balance data, metadata capture and export within FreeMoCap, and evaluation. No funds are requested for recruitment, new recording equipment, model training, travel, or clinical services. The accompanying PHS 398 Form Page 4 presents the direct costs.

### Personnel

**Principal Investigator Aaron T. Cherian.** Requested support is 3.51 calendar months, equivalent to 29.23% effort over 12 months. At a proposed annual salary base of $60,000, salary requested is $17,538. Provisional fringe at 20% adds $3,508, for a total of $21,046. The PI will carry out harmonization, metadata architecture, export implementation, automated checks, reuse demonstrations, and documentation. He will lead DAPR coordination, evaluation, manuscript preparation, and follow-on research planning.

**Mentor Jonathan Samir Matthis, PhD.** Matthis is President and CEO of the FreeMoCap Foundation. Requested support is 1.17 calendar months, equivalent to 9.74% effort over 12 months. At a proposed annual salary base of $80,000, salary requested is $7,795. Provisional fringe at 20% adds $1,559, for a total of $9,354. He will provide scientific and technical mentorship, review definitions and mappings, advise on architecture and evaluation, and review representative exports and documentation. Monthly meetings and quarterly checkpoints will also support manuscript and follow-on proposal development.

These two personnel lines total $30,400: $25,333 in salaries and $5,067 in provisional fringe. The PI's development effort and Matthis's mentorship will span harmonization, implementation, evaluation, and release.

### Fringe Benefits and Compensation Assumptions

The proposed 20% fringe rate comprises employer Social Security and Medicare contributions at 7.65%, an estimated health-insurance contribution of 9.35%, and an estimated retirement contribution of 3.00%, applied to requested salaries. Health and retirement contributions are planning estimates for benefits to be provided during the award.

The budget assumes employee compensation for the funded work. Salary bases, employment arrangements, and benefit contributions require Foundation confirmation before submission; the proposed rate does not represent an established benefits policy. Charges will reflect allowable costs under the arrangements actually implemented. If either role remains contracted, its costs will be classified appropriately using an agreed scope and rate. The PI's grant effort is separate from the requirement that at least 25% of the PI's institutional appointment involve research.

### Other Direct Costs

**Project storage and backup $600.** Twelve months at a proposed average of $50 per month will support working copies, backup, and retained release candidates for the selected numerical data and derived products. These costs support source preservation, reproducibility, and data management. Final capacity and procurement will be matched to the frozen source inventory; public repository deposition will use available no-cost options where appropriate.

### Indirect Costs and Budget Control

Indirect costs are calculated at 8% of $31,000 in direct costs, yielding $2,480 and a total request of $33,480. This calculation assumes the full direct-cost amount is an allowable base under the final agreement; the signing official will confirm the applicable base. There are no outgoing subawards in this proposed budget. Any change in personnel classification, approved rates, or allowable base will be reflected consistently in the detailed budget, this justification, and the consortium-intent letter before submission.

The proposed direct costs will be reviewed at the six-month report and reconciled at close-out. No essential unconfirmed in-kind contribution is used to balance this budget. The project scope will remain limited to the existing gait and balance datasets, a defined set of movement-data types, metadata capture and export, and representative reuse demonstrations.

**FreeMoCap Foundation, Inc.**
A 501(c)(3) nonprofit organization

15 October 2026

DAPR Pilot Grant Program
University of Southern California

**Re: Harmonizing FreeMoCap Movement Data for Rehabilitation Research**

Dear DAPR Pilot Grant Review Committee:

On behalf of the FreeMoCap Foundation, I confirm the Foundation's support for this application and its willingness to enter into a formal subaward or consortium agreement with the University of Southern California if the project is selected for funding.

**Principal Investigator:** Aaron T. Cherian
**Collaborating institutions:** FreeMoCap Foundation and University of Southern California
**Period of support:** 12 months from the approved project start date

| Direct costs | Indirect costs | Total request |
|---|---|---|
| $31,000 | $2,480 | $33,480 |

The proposed project will harmonize existing FreeMoCap gait and balance validation datasets and incorporate metadata capture and harmonized exports into FreeMoCap. The Foundation will provide administrative oversight for the approved scope of work, manage project funds, and meet applicable reporting, research, data-management, and sharing requirements.

If selected, the Foundation will work with USC to finalize the scientific, administrative, and financial arrangements and complete the formal agreement and any required assurances or certifications before award.

Sincerely,

**Jonathan Samir Matthis**
President and CEO, FreeMoCap Foundation
jon@freemocap.org

**Foundation Resource and Stewardship Support**

[PLACE ON FREEMOCAP FOUNDATION LETTERHEAD AFTER APPROVAL]

[DATE]

Dear DAPR Pilot Grant Review Committee,

I am writing on behalf of the FreeMoCap Foundation in support of the proposed Data STACHE project, “Harmonizing FreeMoCap Movement Data for Rehabilitation Research,” led by Aaron T. Cherian, [CONFIRMED CREDENTIALS AND APPOINTMENT].

The Foundation supports making quantitative movement measurement accessible through free and open-source software. As researchers use FreeMoCap in different studies and settings, consistent documentation and interpretable outputs become increasingly important. The proposed project will harmonize existing gait and balance data with measurement definitions, processing history, and DAPR terminology, informed by existing standards. Building the required metadata capture and export capability into FreeMoCap will make that representation available to other researchers. Evaluation using the validation datasets will establish supported uses and identify information needed for broader reuse.

Subject to the applicable permissions and institutional review, the Foundation will support the team’s use of public FreeMoCap validation resources and related technical documentation. Cherian will carry out harmonization, implementation, evaluation, and documentation. Jonathan Samir Matthis, PhD, President and CEO of the Foundation, will provide budgeted mentorship and scientific and technical review. We will designate [NAME AND ROLE] as the organizational contact for repository stewardship and the ordinary software contribution process. Any resource commitments beyond the funded team will require explicit approval.

The Foundation intends to preserve the project’s archived release links, identify a contact for correction requests, and make the documentation discoverable through an appropriate FreeMoCap repository. Proposed metadata and export changes to maintained software will receive ordinary technical review. A documented exporter for the supported inputs will remain available independently of its inclusion in a particular application release.

Matthis’s mentorship will support the PI’s continued development in rehabilitation data science, interpretation of evaluation findings, and preparation of manuscript materials and follow-on research proposals. Additional institutional resource commitments will be documented explicitly and approved through our normal processes.

Sincerely,

[AUTHORIZED OFFICIAL NAME AND TITLE]
FreeMoCap Foundation
[CONTACT INFORMATION]

Signature: _____
Date: _____

**Biosketch Preparation Text**

These narratives are proposed text for Aaron T. Cherian and Jonathan Samir Matthis to review for entry into SciENcv. They do not replace the NIH Biographical Sketch Common Form and Supplement generated through SciENcv, verified by each individual, and combined into the required PDF. Education, appointments, products, identifiers, and certifications must come from each individual's current records.

**Aaron T Cherian Personal Statement**

I am the Chief Scientific Officer of the FreeMoCap Foundation, where my work centers on accessible markerless motion capture and the interpretation and validation of biomechanical measurements. I have worked with multi-camera reconstructions, pose-estimation outputs, joint kinematics, gait events, and standing balance measures, including comparisons with marker-based reference data. This experience is directly relevant to the proposed project because effective harmonization requires an understanding of the anatomical and computational choices underlying each measurement.

I will lead scientific design, harmonization, metadata architecture, export implementation, evaluation, DAPR coordination, and development of manuscript materials and follow-on aims. I will first harmonize the existing gait and balance datasets, then implement the required metadata capture and export within FreeMoCap. Jonathan Samir Matthis will provide scientific and technical mentorship and review of definitions, design, and evaluation. This work extends my experience into rehabilitation data standards and semantic interoperability. My goal is to make harmonized exports available to FreeMoCap researchers so that movement data retain the context needed for interpretation and reuse across studies.

**Proposed Contributions to Science Narratives**

**Accessible movement measurement and reproducible analysis.** My work contributes to making quantitative movement analysis available through an open-source multi-camera platform. This includes development and use of analysis workflows that connect reconstructed movement trajectories to interpretable biomechanical outcomes. The relevance to the present project is practical experience with the data structures and processing decisions that must be documented for another researcher to reproduce or reuse a measurement. [INSERT VERIFIED SOFTWARE OR PUBLICATION PRODUCTS AND PRECISE ROLE.]

**Evaluation of markerless gait and balance measurements.** I have worked on comparisons of FreeMoCap outputs with marker-based reference measurements across gait and standing balance tasks. This work has required attention to differences among pose-estimation approaches and to the definitions of trajectories, joint angles, gait events, and center-of-mass measures. It motivates the proposed distinction between representing measurements consistently and establishing that they are interchangeable. [INSERT VERIFIED MANUSCRIPT OR DATASET CITATIONS, CURRENT PUBLICATION STATUS, AND PRECISE ROLE.]

**Adaptation of movement analysis to prosthetic morphology.** My work on prosthetic gait has examined the use of general-purpose and task-specific pose estimation to describe movement under different alignment conditions. The experience highlights why landmark definitions and model provenance are necessary to interpret reconstructed kinematics. It informs my approach to documenting measurement definitions for researchers working with diverse movement patterns. [INSERT VERIFIED PRODUCTS, CURRENT STATUS, AND PRECISE ROLE.]

**Biosketch Preparation Continued**

**Jonathan Samir Matthis Personal Statement**

I am President and CEO of the FreeMoCap Foundation. My role in the proposed project will be to mentor Aaron T. Cherian as he develops a harmonized representation of existing gait and balance data and incorporates it into a reusable FreeMoCap export. I will provide scientific and technical guidance, review measurement definitions and terminology mappings, advise on metadata architecture and evaluation design, and examine representative exports and worked examples. Monthly meetings and quarterly checkpoints will support progress, interpretation of findings, and preparation of manuscript materials and follow-on research aims.

[MATTHIS TO REVIEW AND ADD VERIFIED EXPERIENCE, RELEVANT PRODUCTS, AND PRECISE CONTRIBUTIONS. This draft describes the proposed role; it does not substitute for a complete record of qualifications.]

**Records Needed for the Final SciENcv Documents**

Cherian and Matthis should each provide verified degree history and dates, appointments, ORCID and other required identifiers, current products and citations, and disclosures required by the current SciENcv workflow. Prepare the required biosketches for the PI and funded mentor, with the institution confirming personnel classifications and application requirements. Do not list DAPR advisors as confirmed personnel before an agreement is in place.

The final combined PDF must list the PI first. NIH ESI status must be checked from the PI's actual record; the proposed personal statement does not establish that status or satisfy the RFP's degree and appointment requirements.

### Application Completion Notes

This version supports Aaron T. Cherian as proposed PI and Jonathan Samir Matthis, PhD, President and CEO of the FreeMoCap Foundation, as mentor. The project retains its two sequential aims and the one-page aims, three-page strategy, and one-page dissemination allocation. The application remains a draft requiring approved effort and rates, institutional information, confirmed permissions, and applicant review.

### Personnel and Scope

The separate programmer position has been removed. Cherian will carry out harmonization, metadata architecture, export implementation, checks, evaluation, and documentation. Matthis will provide mentorship, scientific and technical review, and guidance on dissemination and follow-on research. These roles are reflected in the strategy, resources, timeline, budget, letters, and biosketch preparation materials.

Reproducibility evaluation now consists of the PI running documented examples in a clean environment, with mentor review of representative exports, instructions, and results. It does not claim independent replication by a separate operator. Monthly mentoring reviews and quarterly checkpoints will support the work throughout the award.

The budget retains $31,000 direct and $2,480 indirect costs. Its second personnel line now supports Matthis as mentor. The existing 3.51 and 1.17 calendar-month allocations, salary bases, and 20% fringe remain illustrative planning assumptions requiring approval; the role change does not establish actual compensation or agreed effort.

Aim 1 harmonizes the existing gait and balance datasets with DAPR consultation; Aim 2 builds that representation into FreeMoCap metadata capture and export. The preferred opening, measurement examples, and use of existing standards remain. The datasets come from one study, with broader multi-study reuse enabled by the export. The manuscript plan retains a submission window in the year after completion.

### Scientific Decisions to Resolve

| Decision | Required action |
|---|---|
| Program fit | The proposal uses one validation study and reusable export infrastructure to support future multi-study harmonization. It does not claim two independent studies; DAPR acceptance of this scope remains unconfirmed. |
| Initial export scope | Confirm supported FreeMoCap versions, data types, and source formats. Select representative gait and balance analyses and the metadata needed for their use. |
| DAPR and standards | Review the representation and mappings with the Resource Core. Confirm receiving requirements and applicable standards; retain unresolved concepts explicitly. |
| Manuscript plan | Review the proposed paper and post-award window. The call requires a plan to submit at least one journal manuscript. If no submission is intended, seek DAPR clarification before claiming this requirement is met. |
| Effort and implementation | Confirm that harmonization, architecture, export integration, and evaluation fit the PI’s approved effort and the mentor’s review capacity. Keep additional fields and graphical interface enhancements optional. |

## Submission Readiness

| Item | Required action |
|---|---|
| Required LOI | Confirm submission and retain the receipt. The uploaded RFP lists September 25; the website lists September 30. If no timely LOI was submitted, ask DAPR whether a full proposal may be considered. |
| PI eligibility | Verify the degree, full-time appointment at an NIH-eligible US institution, at least 25 percent research allocation, and ESI or new-area eligibility. An officer title or contractor role alone does not establish eligibility. |
| Team and rates | Confirm Cherian's PI effort and Matthis's mentorship effort, availability, and personnel classifications. Replace illustrative salary bases and 20% fringe with approved rates. The draft retains 3.51 and 1.17 calendar months, $31,000 direct, and $2,480 indirect. |
| Data readiness | Verify the complete validation archive and freeze its version. Its DOI is cited from repository documentation. Confirm the associated records, analysis code, and permissions. Add any verified validation manuscript citation and its current status. |
| Human subjects | Confirm the classification and documentation for the proposed secondary use. The original study had Northeastern IRB approval and the PI retains access to participant identities. Use an accurate study-specific justification or the applicable NIH human-subjects documents. |
| Institutional agreement | Enter dates and official contact details, obtain institutional approval, and sign the consortium-intent letter on letterhead. No approval or signature is represented as obtained. |
| Biosketches and support | Prepare verified SciENcv Common Forms and Supplements for Cherian and Matthis as required. Use the new-area justification only if applicable; include support letters after approval and signature. |

## Submission Assembly

The deadline is October 16, 2026, at 5 PM Pacific or 8 PM Eastern. The portal is https://usc.qualtrics.com/jfe/form/SV_dpCgyqNk9FUNdhY. This package has not been submitted. Confirm the current upload fields and use the individual attachments requested by the portal.

Sections 01-10 contain the scientific and administrative application drafts. Section 03 is the five-page research plan. Sections 07 and 09 are editable Typst working transcriptions of the data-sharing plan and budget; the original filled NIH forms are included in the forms folder. Section 11 is the required institutional intent-letter draft for an external applicant. Section 12 is conditional and section 13 is an optional support-letter draft. Sections 14 and 15 are preparation materials and do not replace certified submission documents.

Edit the Typst section files, finalize placeholders, compile new PDFs, and recheck page and line limits. The combined document is a review copy. Confirm the required form formats when preparing the final attachments; retained Word forms are not automatically synchronized with Typst edits. Revisit source references and affected sections if the data or software scope changes.